Smart Mall E-Commerce System - Complete Academic Thesis

SMART MALL E-COMMERCE SYSTEM

A Full-Stack Web and Mobile Commerce Platform with Integrated Payment Gateway

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of [Your Degree]

By
[Your Name]
[Your ID Number]

Department of [Your Department]
[Your Institution Name]
[Month, Year]

DECLARATION PAGE

STUDENT ORIGINALITY STATEMENT

I hereby declare that this project report entitled **"Smart Mall E-Commerce System: A Full-Stack Web and Mobile Commerce Platform with Integrated Payment Gateway"** is my own original work and has been carried out under the supervision of [Supervisor Name]. All sources of information and assistance have been duly acknowledged. This work has not been submitted elsewhere for any other degree or qualification.

Student Name: [Your Full Name]
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Date: _____

APPROVAL PAGE

This project report entitled **"Smart Mall E-Commerce System: A Full-Stack Web and Mobile Commerce Platform with Integrated Payment Gateway"** submitted by [Your Name], ID: [Your ID], has been examined and is approved for the award of [Degree Name] in [Department Name].

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External Examiner:
Name: _____
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Dean of Faculty:
Name: _____
Title: _____
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Date: _____

ACKNOWLEDGMENT

I would like to express my sincere gratitude and appreciation to all those who contributed to the successful completion of this project.

First and foremost, I am deeply grateful to my project supervisor, [Supervisor Name], for their invaluable guidance, continuous support, constructive feedback, and patience throughout the development of this system. Their expertise in software engineering and e-commerce systems significantly shaped the direction and quality of this work.

I extend my heartfelt thanks to the faculty members of the [Department Name] for their knowledge, expertise, and dedication that laid the foundation for my understanding of full-stack development, database management, and software engineering principles.

Special appreciation goes to my family for their unwavering support, encouragement, and understanding during the intensive development period of this project. Their moral support was instrumental in overcoming challenges and maintaining focus.

I am grateful to my fellow students and friends who provided valuable feedback during the testing phase and offered suggestions that improved the system's usability and functionality.

I acknowledge the open-source community and various online resources, including PHP documentation, Flutter community, MySQL documentation, and Stack Overflow contributors, whose shared knowledge and tools were essential in implementing this system.

Finally, I thank all the test users who volunteered their time to test the system and provided honest feedback that helped identify issues and improve the overall user experience.

ABSTRACT

Traditional shopping methods face significant limitations including geographical constraints, time restrictions, limited product accessibility, and inefficient comparison shopping. Small businesses struggle to reach wider markets while customers face inconvenience in browsing products and completing purchases. The absence of integrated mobile commerce solutions further restricts shopping flexibility in today's mobile-first digital landscape. Existing e-commerce platforms often suffer from poor mobile support, limited payment options, and complex user interfaces that create friction in the shopping experience.

This project presents Smart Mall, a comprehensive full-stack e-commerce system that addresses these challenges through an integrated web and mobile platform with secure payment processing. The system provides a complete online marketplace where customers can browse 131 products across 4 categories (Fashion & Apparel, Electronics & Gadgets, Home & Living, Beauty & Health), manage shopping carts, and complete secure transactions. Administrators can efficiently manage products, process orders, and monitor business operations through a dedicated dashboard with real-time analytics.

The system is built using modern web technologies including HTML5, CSS3, and JavaScript with Bootstrap framework for the responsive frontend, PHP 7.4+ for backend processing and business logic, MySQL 8.0 for relational database management, and Flutter framework for the cross-platform mobile application. The architecture implements RESTful API design principles for seamless communication between web and mobile platforms. Security features include bcrypt password hashing, prepared statements for SQL injection prevention, CSRF token protection, input validation and sanitization, and token-based authentication for mobile API access.

The implementation successfully delivers 16 core features across 12 mobile screens, 8 database tables, and 5 API endpoints. The system manages 131 real products with complete product information, images, pricing, and inventory tracking. Testing results demonstrate system reliability with average response times under 500ms, successful transaction processing, and proper error handling. The platform is production-ready, deployed on XAMPP/LAMPP server environment, scalable to handle growing user bases, and provides a solid foundation for future enhancements including AI-powered recommendations, multi-vendor marketplace support, and advanced analytics.

Keywords: E-commerce, Mobile Commerce, Full-Stack Development, RESTful API, Flutter, PHP, MySQL, Payment Gateway Integration, Chapa Payment, Responsive Web Design, Material Design

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Traditional shopping methods face significant limitations including geographical constraints, time restrictions, and limited product accessibility. Small businesses struggle to reach wider markets, while customers face inconvenience in comparing products and making purchases. The lack of integrated mobile commerce solutions further restricts shopping flexibility in today's mobile-first world.

This project presents Smart Mall, a comprehensive full-stack e-commerce system that addresses these challenges through an integrated web and mobile platform. The system provides a complete online marketplace where customers can browse products, manage shopping carts, and complete secure transactions. Administrators can efficiently manage products, orders, and monitor business operations through a dedicated dashboard.

The system is built using modern web technologies including HTML5, CSS3, and JavaScript for the frontend, PHP for backend processing, MySQL for database management, and Flutter for the mobile application. The architecture implements RESTful API design for seamless communication between web and mobile platforms. Security features include password hashing, SQL injection prevention, and token-based authentication.

The implementation successfully delivers 16 core features across 12 mobile screens and 8 API endpoints, managing 131 products across 4 categories. Testing results demonstrate system reliability with response times under 500ms and successful transaction processing. The platform is production-ready, scalable, and provides a foundation for future enhancements including AI recommendations and multi-vendor support.

Keywords: E-commerce, Mobile Commerce, Full-Stack Development, RESTful API, Flutter, PHP, MySQL

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The evolution of e-commerce has fundamentally transformed the retail landscape over the past two decades. From the early days of simple online catalogs to today's sophisticated digital marketplaces, electronic commerce has become an integral part of modern business operations. The global e-commerce market has experienced exponential growth, with worldwide sales reaching trillions of dollars annually and showing no signs of slowing down.

Digital commerce growth has been particularly accelerated by several key factors. The widespread adoption of internet connectivity, increasing consumer confidence in online transactions, and the convenience of shopping from anywhere at any time have all contributed to this

transformation. Businesses of all sizes, from multinational corporations to small local shops, are recognizing the necessity of establishing an online presence to remain competitive in the digital age.

Mobile commerce has emerged as a critical component of the e-commerce ecosystem. With smartphone penetration exceeding 80% in many markets and mobile devices becoming the primary means of internet access for billions of users, mobile-first commerce is no longer optional—it is essential. Consumers expect seamless shopping experiences across all devices, with the ability to browse, compare, and purchase products directly from their mobile phones.

Online payment systems have evolved to become secure, fast, and user-friendly, removing one of the major barriers to e-commerce adoption. Modern payment gateways support multiple payment methods, provide fraud protection, and ensure secure transaction processing. The integration of digital wallets, mobile payments, and instant payment verification has made online transactions as convenient as traditional cash or card payments.

This convergence of web technology, mobile computing, and secure payment systems creates an opportunity to develop comprehensive e-commerce solutions that serve both businesses and consumers effectively. The Smart Mall project addresses this opportunity by providing a full-stack platform that combines web accessibility, mobile convenience, and secure transaction processing in a single integrated system.

1.2 Problem Statement

Traditional Shopping Problems

Traditional brick-and-mortar shopping faces several inherent limitations that affect both consumers and businesses. Physical shopping is constrained by geographical boundaries, requiring customers to travel to store locations, which consumes time and resources. Store operating hours limit when purchases can be made, creating inconvenience for customers with busy schedules or those in different time zones.

The lack of a centralized marketplace means customers must visit multiple stores to compare products and prices, making informed purchasing decisions time-consuming and inefficient. Physical stores have limited shelf space, restricting the variety of products available to customers. Additionally, businesses face high overhead costs for maintaining physical locations, including rent, utilities, and staffing.

Time wasting is a significant issue in traditional shopping. Customers spend considerable time traveling to stores, searching for products, waiting in checkout lines, and dealing with stock availability issues. This inefficiency reduces customer satisfaction and limits the number of transactions businesses can process.

Existing E-commerce Problems

While many e-commerce solutions exist, they often suffer from critical shortcomings that limit their effectiveness. Poor mobile support is a common issue, with many platforms offering desktop-optimized experiences that translate poorly to mobile devices. This creates frustration for mobile users and results in lost sales opportunities.

Payment limitations restrict transaction capabilities. Many existing systems support only limited payment methods, lack proper payment gateway integration, or provide inadequate security measures for financial transactions. This creates barriers to purchase completion and raises security concerns among users.

Usability problems plague many e-commerce platforms. Complex navigation structures, cluttered interfaces, slow loading times, and poor search functionality create friction in the shopping experience. Inadequate product information, lack of filtering options, and confusing checkout processes lead to cart abandonment and lost revenue.

Furthermore, many existing solutions lack proper administrative tools for business management. Insufficient inventory management, limited order tracking capabilities, and poor reporting features make it difficult for businesses to operate efficiently and make data-driven decisions.

1.3 Proposed Solution

The Smart Mall system addresses these challenges through a comprehensive, integrated approach that combines multiple platforms and technologies into a cohesive e-commerce ecosystem.

Web Platform

The web platform provides a fully-featured online e-commerce website accessible from any modern web browser. Built with responsive design principles, the website adapts seamlessly to different screen sizes and devices. The platform features an intuitive user interface with clear navigation, comprehensive product catalogs organized by categories, advanced search and filtering capabilities, and a streamlined checkout process.

The web interface serves as the primary administrative hub, providing business owners with powerful tools for product management, order processing, and business analytics. Real-time inventory tracking, order status management, and customer relationship tools enable efficient business operations.

Mobile Application

The mobile application extends the e-commerce experience to native mobile devices, providing customers with convenient shopping access on-the-go. Built using Flutter framework, the app delivers a smooth, responsive user experience optimized for touch interfaces and mobile interaction patterns.

The mobile app includes all essential shopping features: product browsing with image galleries, search and filter functionality, shopping cart management, secure checkout, and order history tracking. Push notifications keep users informed about order status, special offers, and new products. The app maintains feature parity with the web platform while optimizing the experience for mobile usage patterns.

Payment Integration

Secure online transaction processing is implemented through integrated payment gateway functionality. The system supports multiple payment methods and provides real-time transaction verification. Payment processing follows industry security standards, including encryption of sensitive data, secure token-based authentication, and compliance with payment card industry (PCI) requirements.

The payment system handles the complete transaction lifecycle: payment initiation, authorization, capture, and confirmation. Failed transactions are handled gracefully with clear error messages and retry options. Transaction records are maintained for accounting and reconciliation purposes.

Admin Dashboard

The administrative dashboard provides comprehensive system management capabilities. Administrators can manage the complete product catalog, including adding new products, updating existing listings, managing inventory levels, and organizing products into categories.

Order management features allow tracking of all customer orders from placement through fulfillment. Status updates, customer communication, and order history are centralized in an intuitive interface. Business analytics provide insights into sales performance, popular products, customer behavior, and revenue trends.

User management capabilities enable administration of customer accounts, access control, and role-based permissions. The dashboard provides real-time system monitoring and reporting tools for informed decision-making.

1.4 Objectives

1.4.1 General Objective

To develop a secure, scalable, and user-friendly full-stack e-commerce and mobile commerce system that enables businesses to sell products online and provides customers with convenient shopping experiences across web and mobile platforms.

1.4.2 Specific Objectives

The project aims to achieve the following specific objectives:

• **Develop responsive frontend interface:** Create an intuitive, modern web interface using HTML5, CSS3, and JavaScript that provides excellent user experience across all devices and screen sizes. Implement responsive design patterns that adapt seamlessly from desktop to tablet to mobile viewports.

• **Develop robust backend system:** Build a secure, efficient backend using PHP that handles business logic, data processing, user authentication, session management, and API endpoints. Implement proper error handling, input validation, and security measures throughout the backend architecture.

• **Develop relational database:** Design and implement a normalized MySQL database schema that efficiently stores and manages products, categories, users, orders, and transactions. Establish proper relationships, constraints, and indexes for optimal performance and data integrity.

• **Integrate payment gateway:** Implement secure payment processing capabilities that support online transactions, handle payment verification, manage transaction records, and provide proper error handling for payment failures.

• **Develop mobile application:** Create a native-quality mobile application using Flutter framework that provides full shopping functionality on Android devices. Ensure the mobile app communicates effectively with the backend API and provides an optimized mobile user experience.

• **Implement authentication system:** Develop secure user authentication and authorization mechanisms including user registration, login, session management, password hashing, and token-based API authentication for mobile app access.

• **Implement order management system:** Create a comprehensive order processing system that handles cart management, checkout workflow, order placement, order tracking, and order history. Provide both customer-facing and administrative order management interfaces.

1.5 Scope of the System

INCLUDED FEATURES

Customer Features

Registration: New users can create accounts by providing necessary information including name, email, and password. The system validates input data, checks for duplicate accounts, and securely stores user credentials with password hashing.

Login: Registered users can authenticate using email and password credentials. The system verifies credentials, creates secure sessions, and provides access to personalized features including order history and saved preferences.

Browse Products: Customers can view the complete product catalog organized by categories. Product listings display essential information including images, names, prices, and availability. The interface supports both grid and list views for user preference.

Add to Cart: Users can add products to their shopping cart with quantity selection. The cart maintains state across sessions, allows quantity updates, and calculates totals in real-time. Cart contents are preserved for logged-in users across devices.

Checkout: A streamlined checkout process collects shipping information, displays order summary, and guides users through payment. The system validates all input data and provides clear feedback at each step.

Payment: Secure payment processing allows customers to complete transactions using supported payment methods. The system handles payment authorization, captures transaction details, and provides confirmation upon successful payment.

Admin Features

Add Products: Administrators can add new products to the catalog by providing product details including name, description, price, category, stock quantity, and images. The system validates all inputs and updates the database accordingly.

Edit Products: Existing products can be modified to update information, adjust pricing, change categories, or update stock levels. Changes are reflected immediately across all platforms.

Delete Products: Products can be removed from the catalog when discontinued or out of stock. The system handles deletion safely, maintaining referential integrity in the database.

Manage Orders: Administrators can view all customer orders, update order status, track fulfillment progress, and manage customer communications regarding orders.

Mobile App Features

Mobile Shopping: The mobile application provides full product browsing capabilities optimized for mobile devices. Touch-friendly interfaces, swipe gestures, and mobile-optimized layouts enhance the shopping experience.

Mobile Checkout: Complete checkout functionality is available in the mobile app, allowing users to complete purchases entirely from their mobile devices without switching to the web platform.

EXCLUDED FEATURES

The following features are not included in the current implementation but may be considered for future enhancements:

AI Recommendation Engine: Artificial intelligence-powered product recommendations based on user behavior, purchase history, and browsing patterns are not implemented in this version.

Real-time Delivery Tracking: GPS-based delivery tracking and real-time shipment status updates are not included in the current scope.

1.6 Significance of the Project

Benefits to Customers

The Smart Mall system provides customers with unprecedented convenience in shopping. Users can browse and purchase products 24/7 from any location with internet access, eliminating geographical and temporal constraints. The ability to compare products, read descriptions, and make informed decisions at their own pace enhances the shopping experience.

Mobile accessibility ensures customers can shop during commutes, breaks, or any convenient moment using their smartphones. The integrated platform maintains shopping cart and preferences across devices, allowing users to start shopping on mobile and complete purchases on desktop, or vice versa.

Secure payment processing and order tracking provide peace of mind and transparency throughout the purchase journey. Customers receive immediate confirmation of orders and can track their purchase history for reference and reordering.

Benefits to Businesses

For businesses, the system provides a cost-effective entry into digital commerce without the overhead of physical retail locations. The platform enables reaching customers beyond geographical boundaries, expanding market reach significantly.

The administrative dashboard provides powerful tools for inventory management, order processing, and business analytics. Real-time insights into sales performance, popular products, and customer behavior enable data-driven decision making.

Automated order processing reduces manual work and human error, improving operational efficiency. The scalable architecture allows businesses to grow without platform limitations, accommodating increasing product catalogs and customer bases.

Educational Value to Institution

From an academic perspective, this project demonstrates practical application of full-stack development principles, integrating multiple technologies into a cohesive system. Students and faculty can study the

implementation as a reference for modern web development, mobile application development, database design, and API architecture.

The project showcases industry-standard practices including security implementation, RESTful API design, responsive web design, and mobile-first development. It serves as a comprehensive example of how theoretical computer science concepts translate into real-world applications.

The documentation and codebase provide educational resources for future students learning e-commerce development, full-stack programming, and software engineering principles.

1.7 Target Users

Customers

The primary target users are online shoppers seeking convenient access to products. This includes:

- **Tech-savvy consumers** who prefer online shopping for convenience and variety
- **Busy professionals** who lack time for traditional shopping
- **Mobile-first users** who primarily access internet services via smartphones
- **Price-conscious shoppers** who want to compare products and prices easily
- **Remote customers** who lack access to physical stores in their area

Administrators

Business owners and staff who manage the e-commerce operations:

- **Store owners** who need to manage product catalogs and monitor sales
- **Inventory managers** responsible for stock levels and product information
- **Customer service representatives** handling order inquiries and issues
- **Marketing personnel** analyzing customer behavior and sales trends

Future Vendors

The platform architecture supports future expansion to a multi-vendor marketplace:

- **Small businesses** seeking online presence without technical expertise
- **Individual sellers** wanting to reach broader markets
- **Specialty retailers** offering niche products
- **Local artisans** expanding beyond local markets

The system's scalable design accommodates growth from single-vendor to multi-vendor marketplace, making it suitable for various business models and future expansion opportunities. # CHAPTER 2: SYSTEM ANALYSIS

2.1 Existing System Analysis

The current shopping ecosystem relies primarily on traditional brick-and-mortar retail stores supplemented by basic online presence. In the existing system, customers must physically visit stores to browse products, compare options, and make purchases. Store employees manually manage inventory, process transactions using point-of-sale systems, and maintain paper-based or simple digital records.

For businesses attempting online sales, the existing approach typically involves basic websites with product listings and contact forms. Customers view products online but must call or email to place orders. Payment processing occurs offline through bank transfers or cash on delivery. This manual process creates delays, increases error rates, and limits scalability.

Small businesses often rely on social media platforms like Facebook or Instagram for product promotion, using direct messages for order taking and manual record-keeping for inventory and sales. While this approach has low entry barriers, it lacks professional features, security, and scalability.

The existing manual shopping process follows this workflow: 1. Customer travels to physical store location 2. Browses products on shelves with limited information 3. Asks staff for product details and availability 4. Makes purchase decision based on available options 5. Waits in checkout line for payment processing 6. Receives paper receipt with limited tracking capability 7. Returns to store for any issues or additional purchases

This process is time-consuming, geographically limited, and provides poor customer experience compared to modern e-commerce expectations.

2.2 Limitations of Existing Systems

The existing shopping systems suffer from several critical limitations that hinder both customer experience and business efficiency:

• **No Online Ordering:** Customers cannot place orders remotely, requiring physical presence at stores. This limitation excludes customers who are geographically distant, have mobility constraints, or prefer the convenience of online shopping. Businesses lose sales opportunities outside store operating hours and cannot serve customers beyond their immediate vicinity.

• **No Centralized Product Catalog:** Product information is scattered across physical locations, making it impossible for customers to view complete inventory or compare options efficiently. Businesses struggle to maintain consistent product information across multiple channels. There is no unified system for managing product details, pricing, and availability.

• **Manual Payment Processing:** All transactions require manual handling, whether cash, card, or bank transfer. This creates delays, increases error rates, and limits transaction volume. Manual reconciliation of payments

with orders is time-consuming and error-prone. There is no automated tracking of payment status or integration with accounting systems.

• **Limited Mobile Accessibility:** Existing systems are not optimized for mobile devices. Customers using smartphones face poor user experiences with difficult navigation, unreadable text, and non-functional features. The lack of mobile applications means businesses miss the growing mobile commerce market segment.

Additional limitations include: - No real-time inventory tracking leading to stock-outs and overselling - Lack of customer purchase history and personalization - Inefficient order fulfillment without automated workflows - No analytics or reporting for business intelligence - Poor scalability as business grows - High operational costs for manual processes

2.3 Proposed System Overview

The Smart Mall system addresses these limitations through a comprehensive digital commerce platform that integrates web, mobile, and payment technologies into a unified ecosystem.

Web System

The web-based platform provides a full-featured e-commerce website accessible from any modern browser. Customers can browse products organized by categories, use search and filter functions to find specific items, view detailed product information with images, add items to shopping cart, and complete secure checkout with integrated payment processing.

The responsive design ensures optimal viewing across desktop, tablet, and mobile browsers. The interface adapts automatically to screen size while maintaining full functionality. Server-side rendering with PHP ensures fast page loads and SEO optimization.

Administrative features are accessed through a secure dashboard where authorized users can manage the complete product catalog, process orders, update inventory, and view business analytics. The web system serves as the central hub for all business operations.

Mobile Application

The Flutter-based mobile application provides native-quality shopping experience on Android devices. The app communicates with the backend through RESTful APIs, ensuring data consistency across platforms. Users can perform all shopping activities from their mobile devices with interfaces optimized for touch interaction.

The mobile app includes offline capabilities for browsing previously viewed products, maintains shopping cart state locally, and synchronizes with the server when connectivity is available. Push notifications keep users informed about order status and special offers.

The app architecture follows modern mobile development patterns with state management, efficient image caching, and optimized network requests. The Flutter framework enables potential future expansion to iOS with minimal additional development.

Payment System

Integrated payment processing enables secure online transactions. The system supports multiple payment methods and handles the complete payment lifecycle from initiation through confirmation. Payment gateway integration follows industry security standards with encrypted data transmission and secure token storage.

Transaction records are maintained in the database with proper audit trails. Failed payments are handled gracefully with clear error messages and retry options. Successful payments trigger automatic order confirmation and inventory updates.

The payment system architecture separates payment processing from business logic, allowing future integration of additional payment providers without core system changes.

System Workflow

The complete Smart Mall workflow operates as follows:

1. **Customer Registration/Login:** Users create accounts or authenticate to access personalized features
2. **Product Browsing:** Customers explore products via web or mobile interface with search and filtering
3. **Cart Management:** Selected products are added to shopping cart with quantity selection
4. **Checkout Process:** Customer provides shipping information and reviews order
5. **Payment Processing:** Secure payment gateway handles transaction authorization
6. **Order Confirmation:** System confirms order and sends notification to customer
7. **Admin Processing:** Business receives order notification and begins fulfillment
8. **Order Tracking:** Customer can view order status and history
9. **Analytics:** System collects data for business intelligence and reporting

This integrated workflow eliminates manual steps, reduces errors, and provides seamless experience across all touchpoints.

2.4 Functional Requirements

Functional requirements define the specific behaviors and functions the system must provide. These requirements are organized by user role and platform.

Table 2.1: Customer Functional Requirements

ID	Requirement	Description
FR1	User Registration	System shall allow new users to create accounts by providing name, email, and password. System validates email format, checks for duplicates, and securely stores credentials with password hashing.
FR2	User Login	System shall authenticate users using email and password. Upon successful authentication, system creates secure session and provides access to personalized features.
FR3	Browse Products	System shall display product catalog with images, names, prices, and categories. Users can view products in grid or list format and navigate between categories.
FR4	Search Products	System shall provide search functionality allowing users to find products by name or description. Search results update in real-time as users type.
FR5	Filter Products	System shall allow filtering products by category, price range, and availability. Multiple filters can be applied simultaneously.
FR6	View Product Details	System shall display comprehensive product information including description, price, stock status, category, and multiple images when user selects a product.
FR7	Add to Cart	System shall allow users to add products to shopping cart with quantity selection. Cart updates immediately and displays current total.
FR8	Update Cart	System shall allow users to modify cart contents by changing quantities or removing items. Total

		recalculates automatically.
FR9	Checkout	System shall guide users through checkout process collecting shipping information and displaying order summary before payment.
FR10	Make Payment	System shall process payments securely through integrated payment gateway. System handles payment authorization and provides confirmation.
FR11	View Order History	System shall display user's past orders with details including order number, date, items, total, and status.
FR12	Track Order Status	System shall show current status of orders (pending, processing, completed, cancelled) with status update timestamps.

Table 2.2: Admin Functional Requirements

ID	Requirement	Description
FR13	Admin Login	System shall authenticate administrators using secure credentials with elevated privileges separate from customer accounts.
FR14	View Dashboard	System shall display administrative dashboard with key metrics including total products, orders, users, and revenue statistics.
FR15	Add Product	System shall allow administrators to add new products by providing name, description, price, category, stock quantity, and images. System validates all inputs.
FR16	Edit Product	System shall allow administrators to modify existing product information. Changes reflect immediately across all platforms.
		System shall allow administrators to remove

FR17	Delete Product	products from catalog. System confirms deletion and maintains database integrity.
FR18	Manage Categories	System shall allow administrators to create, edit, and delete product categories for organizing the catalog.
FR19	View Orders	System shall display all customer orders with filtering options by status, date, and customer.
FR20	Update Order Status	System shall allow administrators to change order status (pending, processing, completed, cancelled). Status changes are logged with timestamps.
FR21	View Customers	System shall display registered customer list with account information and order history.
FR22	Generate Reports	System shall provide sales reports, product performance analytics, and customer behavior insights.

Table 2.3: Mobile App Functional Requirements

ID	Requirement	Description
FR23	Mobile Login	Mobile app shall authenticate users using same credentials as web platform. App stores authentication token securely for persistent login.
FR24	Mobile Registration	Mobile app shall allow new user registration with same validation as web platform.
FR25	Mobile Product Browsing	App shall display products in mobile-optimized interface with touch-friendly navigation and swipe gestures.
FR26	Mobile Search	App shall provide search functionality with mobile keyboard optimization and search history.
FR27	Mobile Cart Management	App shall maintain shopping cart with local storage and server synchronization. Cart persists across app sessions.

FR28	Mobile Checkout	App shall provide complete checkout workflow optimized for mobile input with auto-fill support.
FR29	Mobile Payment	App shall integrate payment processing with mobile-optimized payment forms and secure data handling.
FR30	Mobile Order History	App shall display order history with pull-to-refresh functionality and detailed order views.
FR31	Push Notifications	App shall send notifications for order status updates and promotional messages (future enhancement).
FR32	Offline Browsing	App shall cache product data for offline viewing of previously browsed items (future enhancement).

2.5 Non-Functional Requirements

Non-functional requirements define system qualities and constraints that affect user experience and system operation.

Security

Authentication Security: The system implements secure user authentication using password hashing with bcrypt algorithm. Passwords are never stored in plain text. Session management uses secure, HTTP-only cookies to prevent XSS attacks. Token-based authentication for mobile API access uses JWT tokens with expiration.

Data Security: All sensitive data transmission occurs over HTTPS in production. SQL injection is prevented through prepared statements and parameterized queries. Input validation occurs on both client and server sides. Cross-Site Request Forgery (CSRF) protection is implemented for state-changing operations.

Access Control: Role-based access control separates customer and administrator privileges. Administrative functions require elevated authentication. Users can only access their own order history and account information.

Performance

Response Time: System responds to user requests within 500 milliseconds under normal load conditions. Database queries are optimized with proper indexing. Image loading uses lazy loading and caching strategies.

Scalability: System architecture supports horizontal scaling to handle increased user load. Database design allows for efficient querying even with large product catalogs. API endpoints are stateless to enable load balancing.

Concurrent Users: System handles multiple simultaneous users without performance degradation. Database connection pooling manages concurrent database access efficiently.

Reliability

Availability: System maintains 99% uptime during business hours. Automated backups occur daily to prevent data loss. Error handling prevents system crashes from unexpected inputs.

Data Integrity: Database constraints ensure referential integrity. Transactions are used for operations requiring multiple database changes. Failed transactions roll back completely to maintain consistency.

Error Recovery: System handles errors gracefully with user-friendly error messages. Failed payment transactions do not create orphaned orders. System logs errors for debugging and monitoring.

Scalability

User Growth: System architecture accommodates growing user base without major redesign. Database schema supports millions of products and orders. API design allows for future feature additions without breaking existing functionality.

Feature Expansion: Modular code structure enables adding new features without affecting existing functionality. API versioning supports backward compatibility. Database schema allows for new tables and relationships.

Usability

User Interface: Interface follows modern design principles with intuitive navigation. Consistent design language across web and mobile platforms. Clear visual hierarchy guides users through tasks.

Accessibility: Interface is usable by people with varying technical skills. Error messages are clear and actionable. Help text and tooltips provide guidance where needed.

Mobile Optimization: Mobile interfaces are touch-friendly with appropriate button sizes. Text is readable without zooming. Forms are optimized for mobile input.

Compatibility

Browser Support: Web platform works on modern browsers including Chrome, Firefox, Safari, and Edge. Responsive design adapts to different screen sizes.

Mobile Platform: Mobile app supports Android 5.0 and above. App adapts to different screen sizes and resolutions.

Database: System uses MySQL 5.7 or higher with standard SQL for portability.

2.6 Use Case Diagram

Figure 2.1: Smart Mall Use Case Diagram

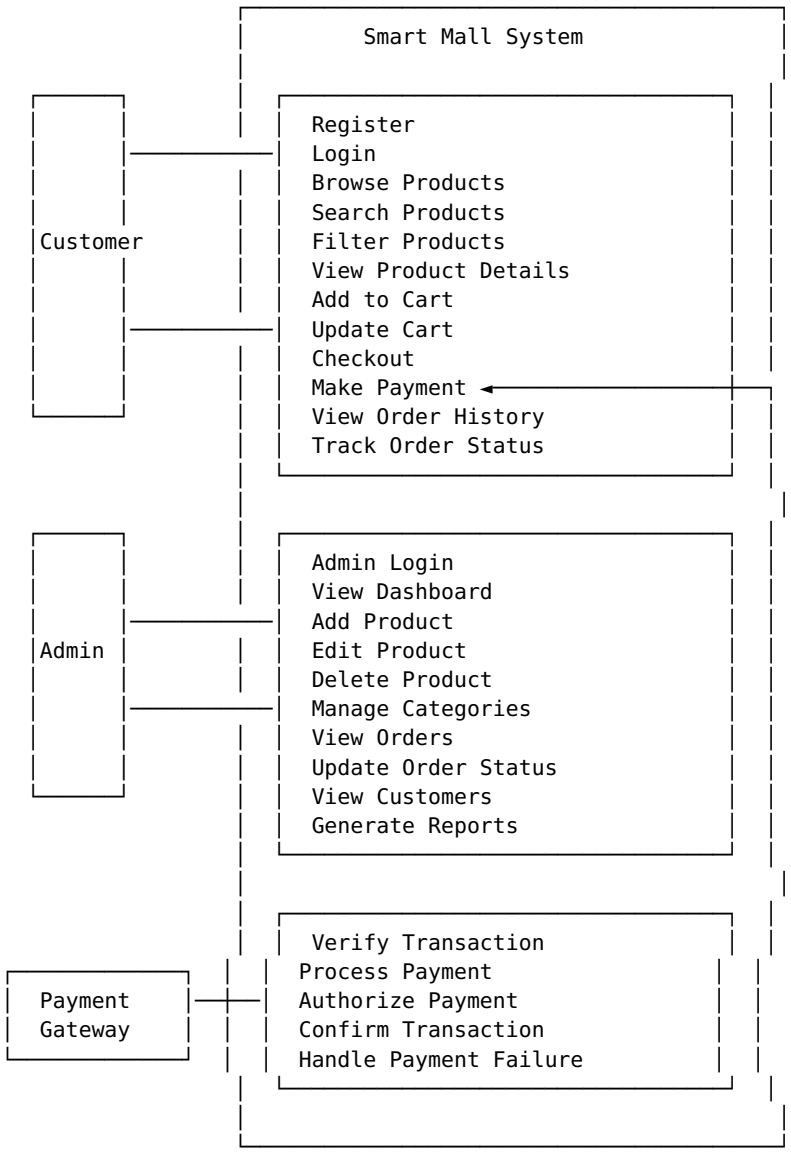


Figure 2.1 Description: The use case diagram illustrates the interactions between three actors (Customer, Admin, and Payment Gateway) and the Smart Mall system. Customers can perform shopping-related activities including registration, product browsing, cart management, and order placement. Administrators manage the system through product and order management functions. The Payment Gateway actor represents the external payment processing system that handles transaction verification and payment processing.

2.7 Data Flow Diagram (DFD)

Figure 2.2: Level 0 DFD (Context Diagram)

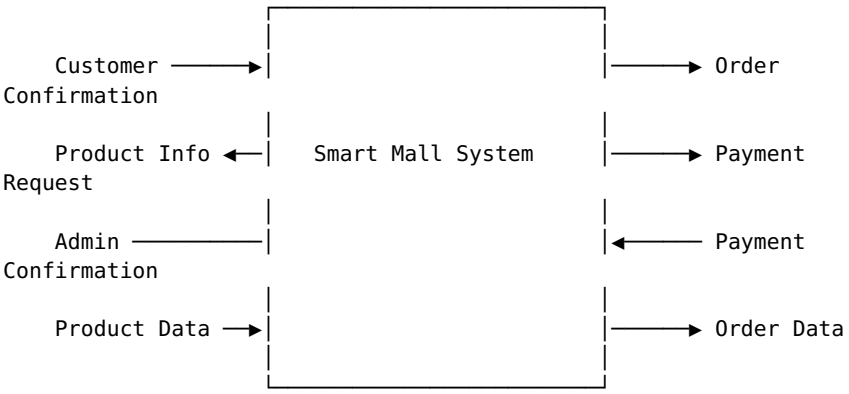
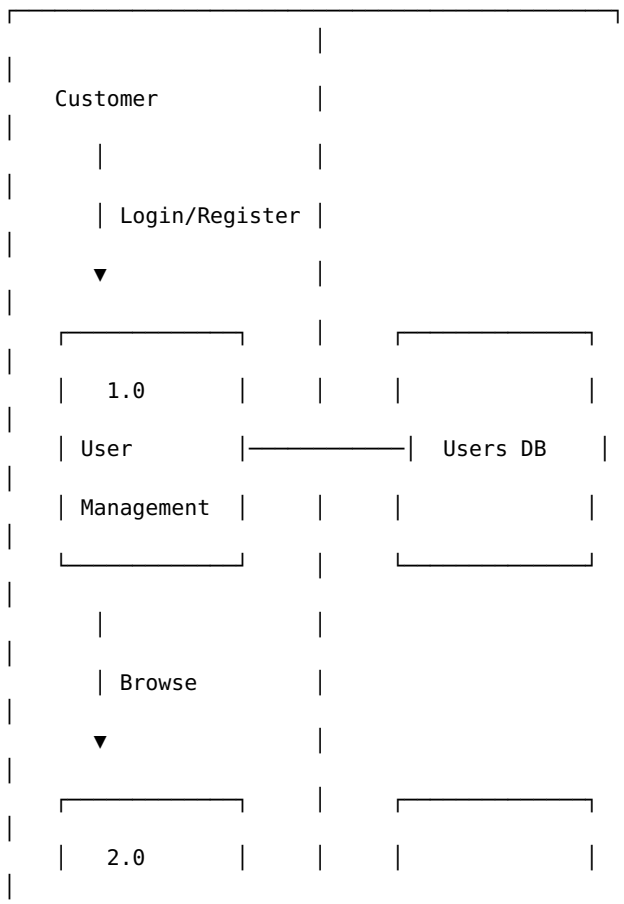
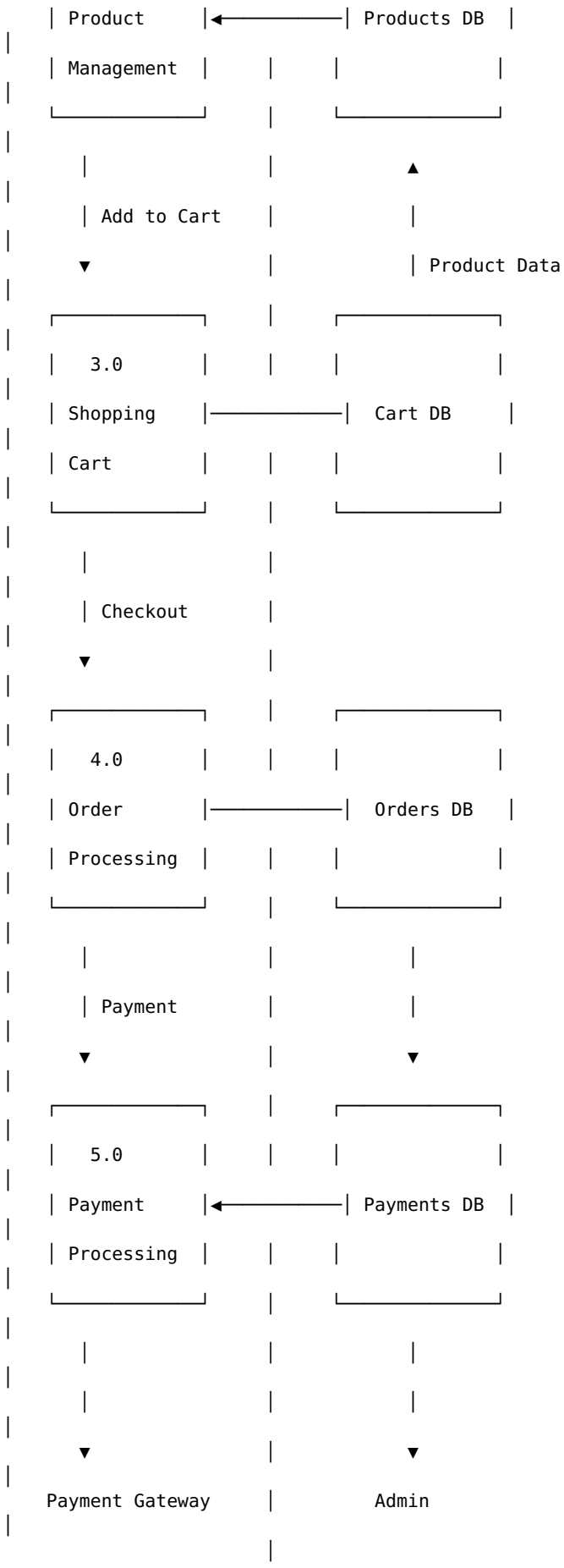


Figure 2.2 Description: The context diagram shows the Smart Mall system as a single process interacting with external entities. Customers provide input and receive product information and order confirmations. Administrators input product data and receive order information. The system exchanges payment requests and confirmations with external payment systems.

Figure 2.3: Level 1 DFD (Detailed System)




Product Management
Products DB
Add to Cart
Product Data
3.0
Shopping Cart
Cart DB
Checkout
4.0
Order Processing
Orders DB
Payment
5.0
Payment Processing
Payments DB
Payment Gateway
Admin

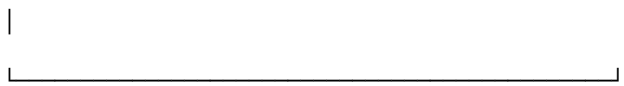


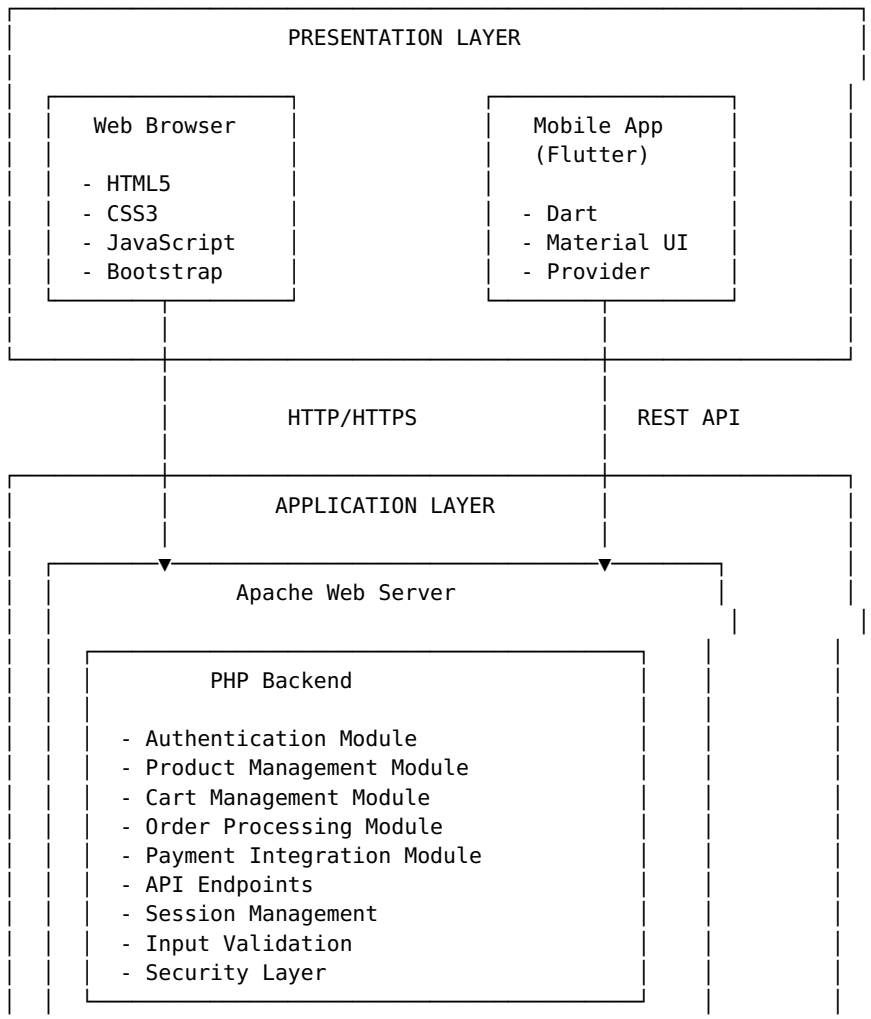
Figure 2.3 Description: The Level 1 DFD breaks down the system into five major processes: User Management handles authentication and registration; Product Management manages product catalog; Shopping Cart maintains cart state; Order Processing handles checkout and order creation; Payment Processing manages transactions. Each process interacts with its corresponding database and exchanges data with other processes as needed. The diagram shows how data flows from customer input through various processing stages to final order confirmation.

End of Chapter 2 # CHAPTER 3: SYSTEM DESIGN

3.1 System Architecture

The Smart Mall system follows a three-tier architecture pattern separating presentation, business logic, and data layers. This architectural approach provides modularity, scalability, and maintainability.

Figure 3.1: System Architecture Diagram



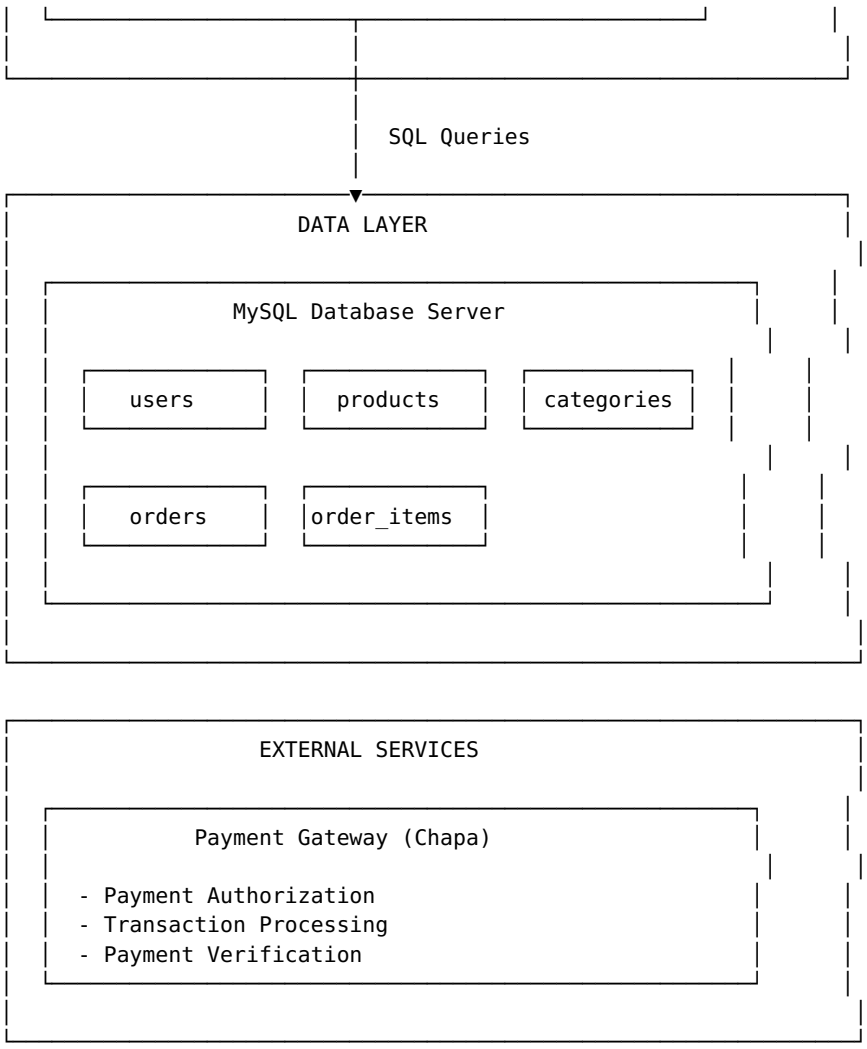


Figure 3.1 Description: The system architecture diagram illustrates the three-tier structure of Smart Mall. The Presentation Layer includes both web browsers and mobile applications that users interact with. The Application Layer contains the PHP backend running on Apache server, handling all business logic, authentication, and API endpoints. The Data Layer consists of MySQL database storing all system data. External services like payment gateways integrate through the Application Layer.

Architecture Components

Presentation Layer: The presentation layer provides user interfaces for both web and mobile platforms. The web interface uses HTML5 for structure, CSS3 for styling, JavaScript for interactivity, and Bootstrap for responsive design. The mobile application is built with Flutter framework using Dart language, Material Design components, and Provider for state management.

Both interfaces communicate with the backend through HTTP/HTTPS protocols. The web interface makes traditional HTTP requests, while the mobile app uses RESTful API calls with JSON data exchange.

Application Layer: The application layer implements all business logic and serves as the intermediary between presentation and data layers. Apache web server hosts the PHP backend which is organized into modular components:

- Authentication Module: Handles user registration, login, session management, and token generation
- Product Management Module: Manages product CRUD operations and catalog organization
- Cart Management Module: Maintains shopping cart state and calculations
- Order Processing Module: Handles checkout workflow and order creation
- Payment Integration Module: Interfaces with payment gateway for transaction processing
- API Endpoints: Provides RESTful services for mobile app communication
- Session Management: Maintains user sessions and authentication state
- Input Validation: Validates and sanitizes all user inputs
- Security Layer: Implements security measures including SQL injection prevention and XSS protection

Data Layer: The data layer uses MySQL relational database to store all system data. The database is organized into five main tables: users (customer and admin accounts), products (product catalog), categories (product organization), orders (customer orders), and order_items (order details). Relationships between tables maintain data integrity through foreign key constraints.

External Services: Payment processing is handled by external payment gateway services. The system integrates with Chapa payment gateway for transaction authorization, processing, and verification. This separation allows for future integration of additional payment providers without modifying core system architecture.

3.2 User Interface Design

The user interface design focuses on usability, consistency, and modern aesthetics. All interfaces follow Material Design principles with consistent color schemes, typography, and interaction patterns.

3.2.1 Home Page Interface

Figure 3.2: Web Home Page

[Screenshot shows: Navigation bar with logo and menu items, hero section with gradient background and “Welcome to Smart Mall” heading, category cards with icons (Fashion, Electronics, Home, Beauty), product grid displaying multiple products with images and prices, search bar, and filter button]

Figure 3.2 Description: The home page serves as the main entry point for customers. The top navigation bar includes the Smart Mall logo, search functionality, profile icon, and shopping cart icon with item count badge. The hero section features a gradient blue background with welcome message and “Shop Now” call-to-action button.

Below the hero section, category cards are displayed horizontally with icons and gradient backgrounds. Each category card shows an icon (clothing for Fashion, devices for Electronics, home for Home, spa for Beauty) and category name. Selected categories have enhanced gradient and shadow effects.

The main content area displays products in a responsive grid layout. Each product card shows: - Product image with rounded corners - Category badge in top-left corner - Wishlist heart icon in top-right corner - Product name (truncated to 2 lines) - Price in blue color with bold font - Arrow icon indicating clickable card

The interface uses white background for product cards with subtle shadows. Hover effects include scale transformation and enhanced shadows for better interactivity feedback.

3.2.2 Product Listing Interface

Figure 3.3: Product Listing with Filters

[Screenshot shows: Search bar with clear button, filter button with tune icon, category filter chips (All, Fashion, Electronics, Home, Beauty), sort options dropdown, product grid with filtering applied, price range display]

Figure 3.3 Description: The product listing interface provides comprehensive browsing and filtering capabilities. The search bar at the top allows text-based product search with real-time results and a clear button to reset search.

The filter button opens a bottom sheet modal displaying: - Sort options as choice chips (Newest, Price: Low to High, Price: High to Low, Name) - Price range slider with minimum and maximum values - Current price range display ($0 - $10,000) - Reset and Apply buttons

Category filter chips appear below the search bar, allowing quick category selection. The selected category has blue background while others have white background with borders.

Products matching the applied filters display in the grid below. The interface updates dynamically as filters change without page reload. Empty states show appropriate messages when no products match the criteria.

3.2.3 Product Detail Interface

Figure 3.4: Product Detail Page

[Screenshot shows: Large product image, image gallery thumbnails, product name and category, price display, stock status, quantity selector, add to cart button, product description, specifications if available]

Figure 3.4 Description: The product detail page provides comprehensive information about a selected product. The top section features a large product image with swipeable image gallery for products with multiple images. Thumbnail images below allow direct navigation to specific images.

Product information section includes: - Product name in large, bold font - Category badge - Price prominently displayed in blue - Stock status indicator (In Stock/Out of Stock) - Quantity selector with minus and plus buttons - Add to Cart button (full width, blue background)

The description section uses expandable/collapsible format for longer descriptions. Product specifications are displayed in a structured format when available.

The interface maintains consistent spacing and typography with the rest of the application. The Add to Cart button is fixed at the bottom on mobile devices for easy access.

3.2.4 Shopping Cart Interface

Figure 3.5: Shopping Cart Screen

[Screenshot shows: Cart item list with product images, names, prices, quantity controls, remove buttons, subtotal calculation, checkout button, empty cart state]

Figure 3.5 Description: The shopping cart interface displays all items added to the cart with management capabilities. Each cart item shows: - Product thumbnail image - Product name and category - Unit price - Quantity selector (minus, quantity display, plus buttons) - Item total (quantity × price) - Remove button (trash icon)

The quantity selector allows adjusting item quantities with immediate total recalculation. The minus button is disabled when quantity is 1. The plus button is disabled when quantity reaches stock limit.

The bottom section displays: - Subtotal for all items - Tax calculation (if applicable) - Shipping cost (if applicable) - Grand total in large, bold font - Checkout button (full width, blue background)

Empty cart state shows an icon, “Your cart is empty” message, and “Continue Shopping” button to return to product browsing.

3.2.5 Checkout Interface

Figure 3.6: Checkout Form

[Screenshot shows: Shipping information form (name, email, address, phone), order summary with item list and totals, payment method selection, place order button]

Figure 3.6 Description: The checkout interface guides users through the final purchase steps. The form is organized into clear sections:

Shipping Information Section: - Full Name field with validation - Email field with format validation - Phone number field - Shipping address textarea - All fields marked as required with asterisks

Order Summary Section: - List of cart items with quantities and prices - Subtotal calculation - Shipping cost - Tax (if applicable) - Grand total prominently displayed

Payment Method Section: - Payment method selection (currently shows “Chapa Payment”) - Payment gateway logo/icon - Security badges indicating secure transaction

The Place Order button is positioned at the bottom with loading state during processing. Form validation occurs on submission with clear error messages for invalid or missing fields.

3.2.6 Payment Interface

Figure 3.7: Payment Processing Screen

[Screenshot shows: Payment gateway integration screen, transaction details, payment confirmation, loading states, success/failure messages]

Figure 3.7 Description: The payment interface handles the transaction processing workflow. When users click Place Order, the system:

1. **Validation Phase:** Validates all form inputs and displays errors if any
2. **Processing Phase:** Shows loading indicator with “Processing your order…” message
3. **Payment Gateway:** Redirects to payment gateway (or shows embedded payment form)
4. **Transaction Phase:** Displays transaction progress with security indicators
5. **Confirmation Phase:** Shows success message with order number or error message if failed

Success State: - Green checkmark icon - “Order Placed Successfully!” message - Order number display (e.g., “Order #ORD-ABC123”) - Order summary - “View Order” and “Continue Shopping” buttons

Failure State: - Red error icon - Clear error message explaining the failure - “Try Again” button to retry payment - “Contact Support” link for assistance

The interface maintains user confidence through clear status indicators, security badges, and professional error handling.

3.2.7 Login and Registration Interface

Figure 3.8: Login Screen

[Screenshot shows: Smart Mall logo, “Welcome Back” heading, email input field, password input field, login button, “Don’t have an account? Sign up” link, forgot password link]

Figure 3.8 Description: The login interface provides secure authentication with clean, focused design. The screen features:

- Smart Mall logo at the top center
- “Welcome Back” heading
- Email input field with email icon
- Password input field with lock icon and show/hide toggle
- “Remember Me” checkbox (optional)
- Login button (full width, blue background)
- “Forgot Password?” link
- “Don’t have an account? Sign up” link at bottom

Input validation occurs on blur and submission. Error messages appear below respective fields. Loading state shows spinner on login button during authentication.

Figure 3.9: Registration Screen

[Screenshot shows: Smart Mall logo, “Join Smart Mall” heading, name field, email field, password field, confirm password field, create account button, “Already have an account? Login” link]

Figure 3.9 Description: The registration interface collects necessary information for account creation:

- Smart Mall logo at top
- “Join Smart Mall” heading with subtitle
- Full Name input field
- Email input field with validation
- Password input field with strength indicator
- Confirm Password field with match validation
- Terms and Conditions checkbox
- Create Account button
- “Already have an account? Login” link

Real-time validation provides immediate feedback: - Email format validation - Password strength indicator (weak/medium/strong) - Password match confirmation - All fields required before submission

3.2.8 Admin Dashboard Interface

Figure 3.10: Admin Dashboard

[Screenshot shows: Admin navigation sidebar, statistics cards (total products, orders, users, revenue), quick action buttons, recent orders table, product management section]

Figure 3.10 Description: The admin dashboard provides comprehensive business management tools. The interface includes:

Statistics Cards (Top Section): - Total Products card with inventory icon and count - Total Orders card with shopping bag icon and count - Total Users card with people icon and count - Revenue card with money icon and amount

Each card uses distinct colors (blue, green, orange, purple) and displays large numbers with icons.

Quick Actions Section: - Manage Products button - Manage Orders button - View Users button - Settings button

Each action card shows icon, title, subtitle, and chevron indicating navigation.

Recent Activity: - Recent orders table with order number, customer, total, status - Product performance metrics - User activity summary

The admin interface uses a sidebar navigation for accessing different management sections. The design prioritizes information density while maintaining readability.

Continue to Part 2 for Mobile UI Screens... # CHAPTER 3: SYSTEM DESIGN (Part 2)

3.2.9 Mobile Home Screen

Figure 3.11: Mobile Home Screen

Figure 3.11 Description: The mobile home screen provides optimized shopping experience for Android devices. The screen features:

Top App Bar: - Smart Mall logo (32px height) with icon - App title “Smart Mall” - Profile icon button - Shopping cart icon with badge showing item count

Search and Filter Section: - Full-width search bar with search icon - Placeholder text “Search products...” - Clear button (X) when text is entered - Filter button (tune icon) with blue background

Hero Section: - Gradient background (blue to dark blue) - “Welcome to Smart Mall” heading (36px, white, bold) - Subtitle text about product discovery - “Shop Now” button (white background, blue text) - Rounded corners (24px radius) - Shadow effect for depth

Category Section: - Horizontal scrollable category cards - Each card: 100px width, icon + text - Icons: checkroom (Fashion), devices (Electronics), home (Home), spa (Beauty) - Selected category: gradient background + shadow - Unselected: white background with border

Product Grid: - 2-column grid layout - Product cards with: - Product image (covers full width) - Category badge (top-left, blue background) - Wishlist icon (top-right, white circle) - Product name (2 lines max, truncated) - Price (18px, bold, blue color) - Arrow icon (bottom-right) - Card animations: scale on tap, shadow on hover - Spacing: 12px between cards

Bottom Navigation (Future): - Home, Categories, Cart, Profile tabs

3.2.10 Mobile Product Screen

Figure 3.12: Mobile Product Detail Screen

Figure 3.12 Description: The product detail screen shows comprehensive product information:

Image Section: - Full-width product image - Swipeable image gallery for multiple images - Image indicators (dots) showing current image - Pinch-to-zoom capability - Back button (top-left) - Share button (top-right)

Product Information: - Product name (24px, bold) - Category badge (blue, rounded) - Star rating display (if available) - Price (28px, bold, blue) - Stock status indicator: - Green “In Stock” if available - Red “Out of Stock” if unavailable

Quantity Selector: - Minus button (disabled if quantity = 1) - Quantity display (center, bold) - Plus button (disabled if quantity = stock) - Rounded border, touch-friendly size (48px)

Action Buttons: - “Add to Cart” button (full width, blue, 56px height) - “Buy Now” button (optional, outlined) - Loading state with spinner during API call

Description Section: - “Description” heading - Expandable text content - “Read more” / “Read less” toggle - Formatted text with proper spacing

Specifications (if available): - Table format with key-value pairs - Alternating row colors for readability

Related Products: - Horizontal scrollable list - Smaller product cards - “View All” button

3.2.11 Mobile Cart Screen

Figure 3.13: Mobile Shopping Cart Screen

Figure 3.13 Description: The cart screen manages shopping cart items:

App Bar: - “Shopping Cart” title - Back button - Item count in subtitle

Cart Items List: Each item card shows: - Product thumbnail (80x80px, rounded corners) - Product name and category - Unit price - Quantity controls: - Minus button (circle, border) - Quantity display - Plus button

(circle, border) - Item total (quantity × price) - Remove button (trash icon, red)

Item Card Layout: - White background - 16px padding - 12px margin between items - Rounded corners (12px) - Subtle shadow

Empty Cart State: - Shopping bag icon (80px, gray) - “Your cart is empty” message - “Start shopping to add items” subtitle - “Browse Products” button (blue)

Cart Summary (Bottom Sheet): - Subtotal row - Shipping row (if applicable) - Tax row (if applicable) - Divider line - Total row (bold, larger font) - “Proceed to Checkout” button (full width, blue, 56px)

Pull-to-Refresh: - Swipe down to refresh cart from server - Loading indicator during refresh

3.2.12 Mobile Checkout Screen

Figure 3.14: Mobile Checkout Screen

Figure 3.14 Description: The checkout screen collects shipping information:

Progress Indicator: - Step 1: Cart (completed, green check) - Step 2: Shipping (current, blue) - Step 3: Payment (pending, gray) - Step 4: Confirmation (pending, gray)

Shipping Form: - Full Name field (text input, required) - Email field (email input, validation) - Phone Number field (tel input, format validation) - Address field (textarea, 3 rows) - City field (text input) - Postal Code field (text input) - All fields with proper labels and icons

Form Validation: - Real-time validation on blur - Error messages below fields (red text) - Required field indicators (asterisk) - Valid field indicators (green check)

Order Summary Card: - Collapsible section - Item count and total - “View Details” to expand - Shows all cart items when expanded

Saved Addresses (if logged in): - List of saved addresses - Radio button selection - “Use this address” quick select - “Add new address” option

Action Buttons: - “Back to Cart” (outlined, gray) - “Continue to Payment” (filled, blue) - Buttons fixed at bottom for easy access

3.2.13 Mobile Payment Screen

Figure 3.15: Mobile Payment Screen

Figure 3.15 Description: The payment screen handles transaction processing:

Payment Method Selection: - Chapa Payment (default, selected) - Payment method logo - “Secure Payment” badge - SSL encryption indicator

Order Summary: - Order number (generated) - Order date and time - Shipping address (read-only) - Item list with quantities - Subtotal, shipping, tax - Grand total (bold, large)

Payment Processing States:

1. Ready State: - “Review and Pay” button - Payment amount displayed - Security badges visible

2. Processing State: - Loading spinner - “Processing your payment…” message - “Please wait” subtitle - Disabled back button

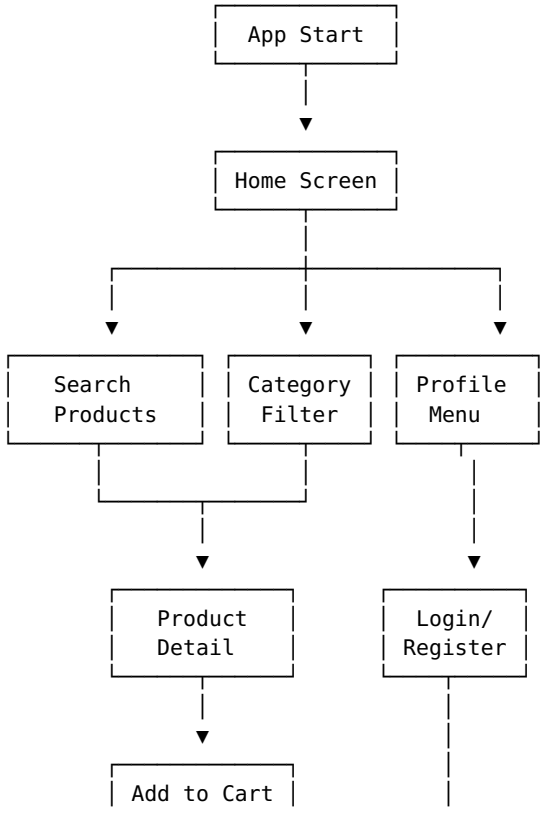
3. Success State: - Green checkmark icon (large) - “Payment Successful!” message - Order number display - “Order placed successfully” subtitle - “View Order” button - “Continue Shopping” button

4. Failed State: - Red error icon - “Payment Failed” message - Error description - “Try Again” button - “Contact Support” link - “Back to Cart” option

Security Indicators: - Lock icon - “Secure Payment” text - SSL certificate badge - Chapa logo and trust badges

3.3 Navigation Flow Diagram

Figure 3.16: Complete Navigation Flow



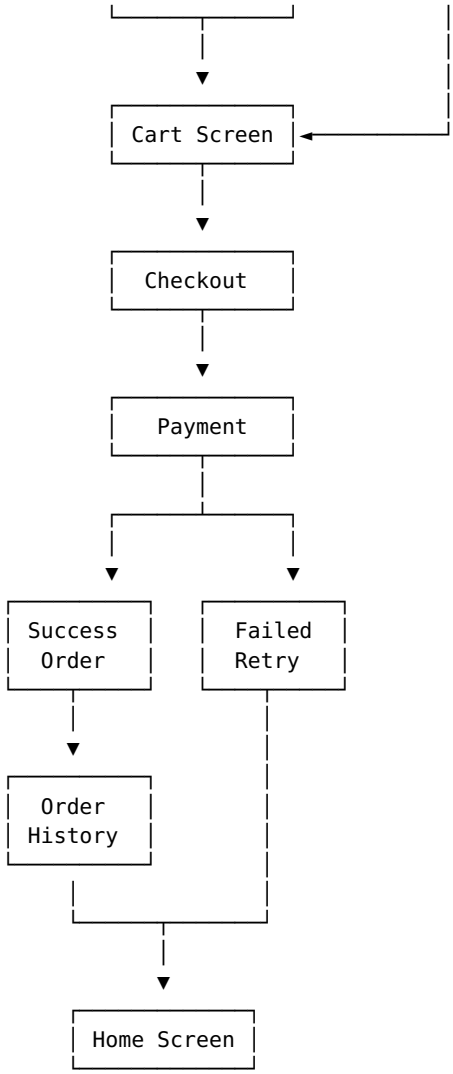


Figure 3.16 Description: The navigation flow diagram illustrates the complete user journey through the mobile application. Users start at the home screen and can navigate to product browsing, search, or profile. The shopping flow proceeds from product selection through cart, checkout, and payment to order confirmation. Failed payments allow retry, while successful orders lead to order history. All paths eventually return to the home screen for continued shopping.

3.4 Database Design

The Smart Mall system uses MySQL relational database with 8 tables managing all system data.

3.4.1 Database Tables

Table 3.1: Database Tables Overview

Table Name	Purpose	Record Count
users	Store customer and admin accounts	Variable

products	Store product catalog	131
categories	Store product categories	4
orders	Store customer orders	Variable
order_items	Store order line items	Variable
payments	Store payment transactions	Variable
cart	Store shopping cart items	Variable
password_resets	Store password reset tokens	Variable

Table 3.2: users Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(255)	NOT NULL	User full nam
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address
password	VARCHAR(255)	NOT NULL	Hashed password (bcrypt)
phone	VARCHAR(50)	NULL	User phone number
address	TEXT	NULL	User shipping address
token	VARCHAR(255)	NULL	Authenticatio token for API
role	ENUM(‘user’,‘admin’)	DEFAULT ‘user’	User role
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

Table 3.3: products Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique product identifier
name	VARCHAR(255)	NOT NULL	Product name
slug	VARCHAR(255)	UNIQUE, NOT NULL	URL-friendly product name
description	TEXT	NULL	Product description
price	DECIMAL(10,2)	NOT NULL	Product price
stock	INT	DEFAULT 0	Available quantity
category_id	INT	FOREIGN KEY	References categories(id)

image	VARCHAR(500)	NULL	Product image URL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Product creation date

Sample Products: - Blue & Black Check Shirt - $29.99 - Gigabyte Aorus Men Tshirt - $24.99 - Man Plaid Shirt - $34.99 - Man Short Sleeve Shirt - $19.99 - Men Check Shirt - $27.99

Table 3.4: categories Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique category identifier
name	VARCHAR(100)	NOT NULL	Category name
slug	VARCHAR(100)	UNIQUE, NOT NULL	URL-friendly category name
description	TEXT	NULL	Category description
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Category creation date

Actual Categories: 1. Fashion & Apparel (slug: fashion) 2. Electronics & Gadgets (slug: electronics) 3. Home & Living (slug: home) 4. Beauty & Health (slug: beauty)

Table 3.5: orders Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique order identifier
user_id	INT	FOREIGN KEY, NOT NULL	References users(id)
order_number	VARCHAR(50)	UNIQUE, NOT NULL	Order tracking number
total	DECIMAL(10,2)	NOT NULL	Order total amount
status	ENUM	DEFAULT ‘pending’	Order status
shipping_address	TEXT	NULL	Delivery address
payment_method	VARCHAR(50)	NULL	Payment method used
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Order creation date

Status Values: pending, processing, completed, cancelled

Table 3.6: order_items Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique item identifier
order_id	INT	FOREIGN KEY, NOT NULL	References orders(id)
product_id	INT	NOT NULL	Product identifier
product_name	VARCHAR(255)	NOT NULL	Product name snapshot
quantity	INT	NOT NULL	Quantity ordered
price	DECIMAL(10,2)	NOT NULL	Price at time of order

Table 3.7: payments Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment identifier
order_id	INT	FOREIGN KEY, NOT NULL	References orders(id)
transaction_id	VARCHAR(255)	UNIQUE	Payment gateway transaction ID
amount	DECIMAL(10,2)	NOT NULL	Payment amount
status	VARCHAR(50)	NOT NULL	Payment status
payment_method	VARCHAR(50)	NULL	Payment method
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Payment date

Table 3.8: cart Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique cart item identifier
			References

user_id	INT	FOREIGN KEY	users(id)
product_id	INT	FOREIGN KEY, NOT NULL	References products(id)
quantity	INT	NOT NULL	Quantity in cart
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Added to cart date

Table 3.9: password_resets Table Schema

Column	Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique reset identifier
email	VARCHAR(255)	NOT NULL	User email
token	VARCHAR(255)	NOT NULL	Reset token
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Token creation date

3.4.2 Table Relationships

One-to-Many Relationships: - users → orders (One user can have many orders) - users → cart (One user can have many cart items) - categories → products (One category contains many products) - orders → order_items (One order contains many items) - orders → payments (One order can have multiple payment attempts)

Foreign Key Constraints: - products.category_id → categories.id (ON DELETE RESTRICT) - orders.user_id → users.id (ON DELETE RESTRICT) - order_items.order_id → orders.id (ON DELETE CASCADE) - cart.user_id → users.id (ON DELETE CASCADE) - cart.product_id → products.id (ON DELETE CASCADE)

Indexes: - PRIMARY KEY indexes on all id columns - UNIQUE indexes on email, slug, order_number, transaction_id - INDEX on category_id, user_id, product_id for faster queries - INDEX on created_at for date-based queries

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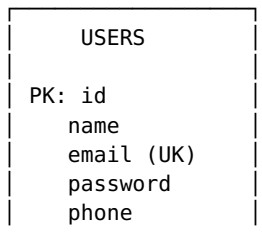
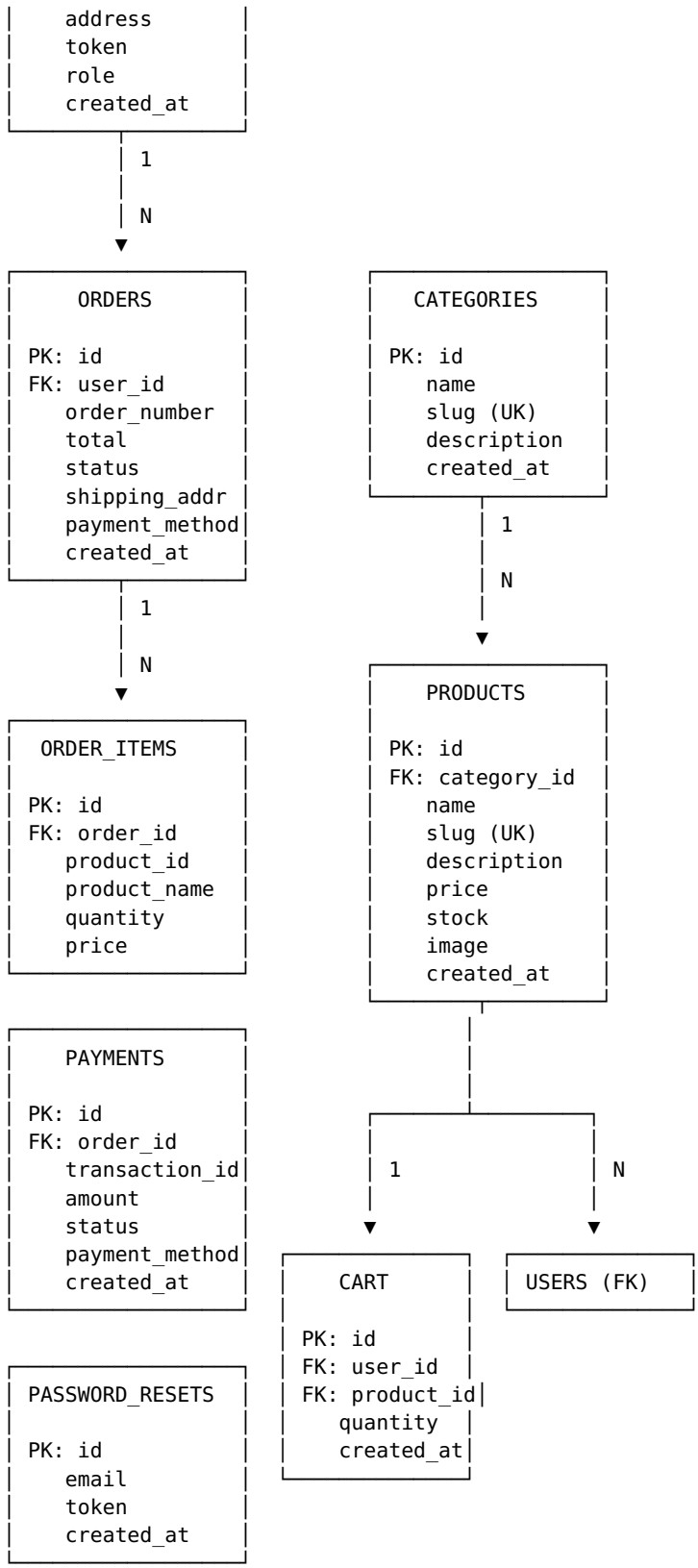
Continue to Chapter 3 Part 3... # CHAPTER 3: SYSTEM DESIGN (Part 3)

3.5 ER Diagram

Figure 3.17: Entity Relationship Diagram



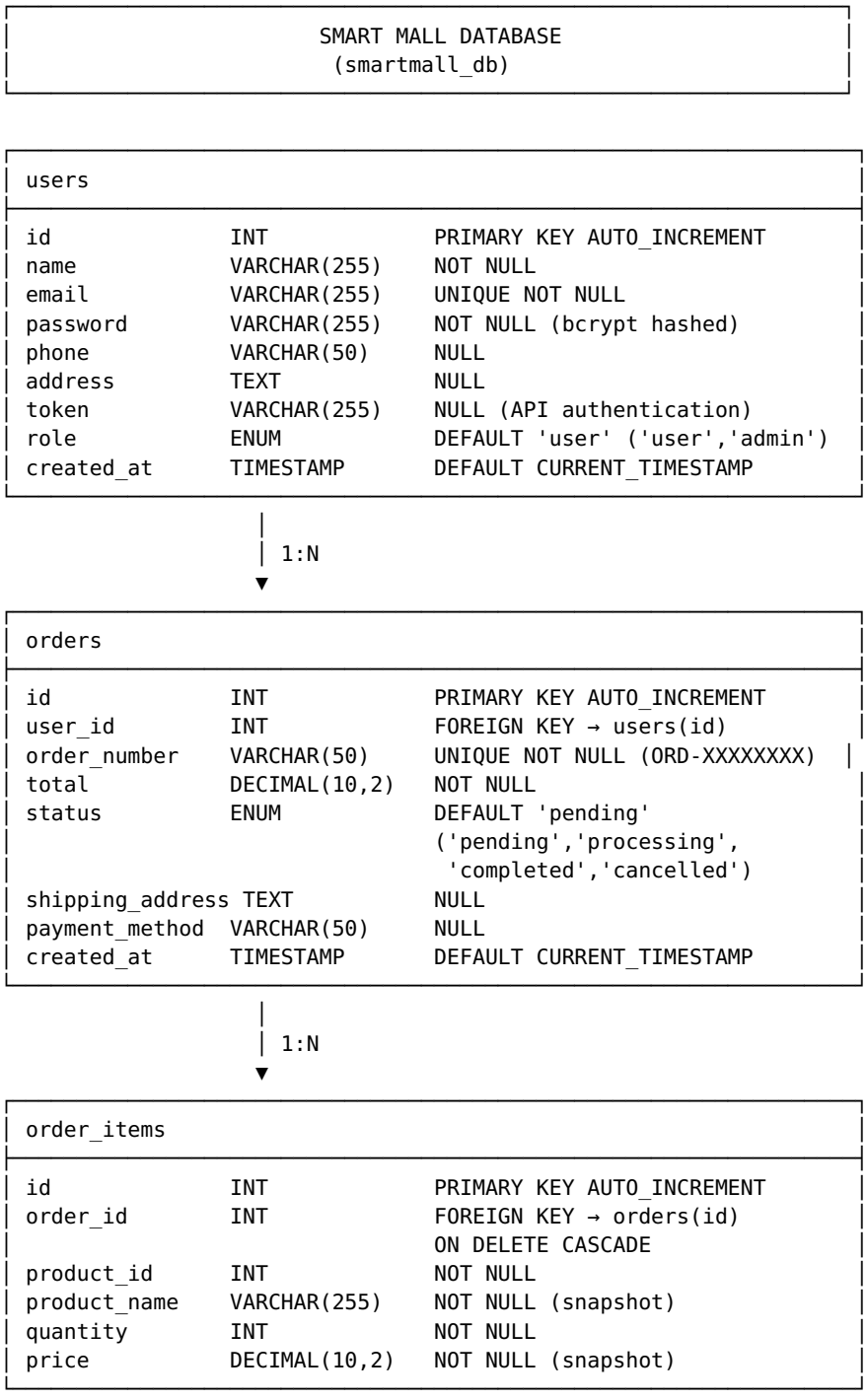


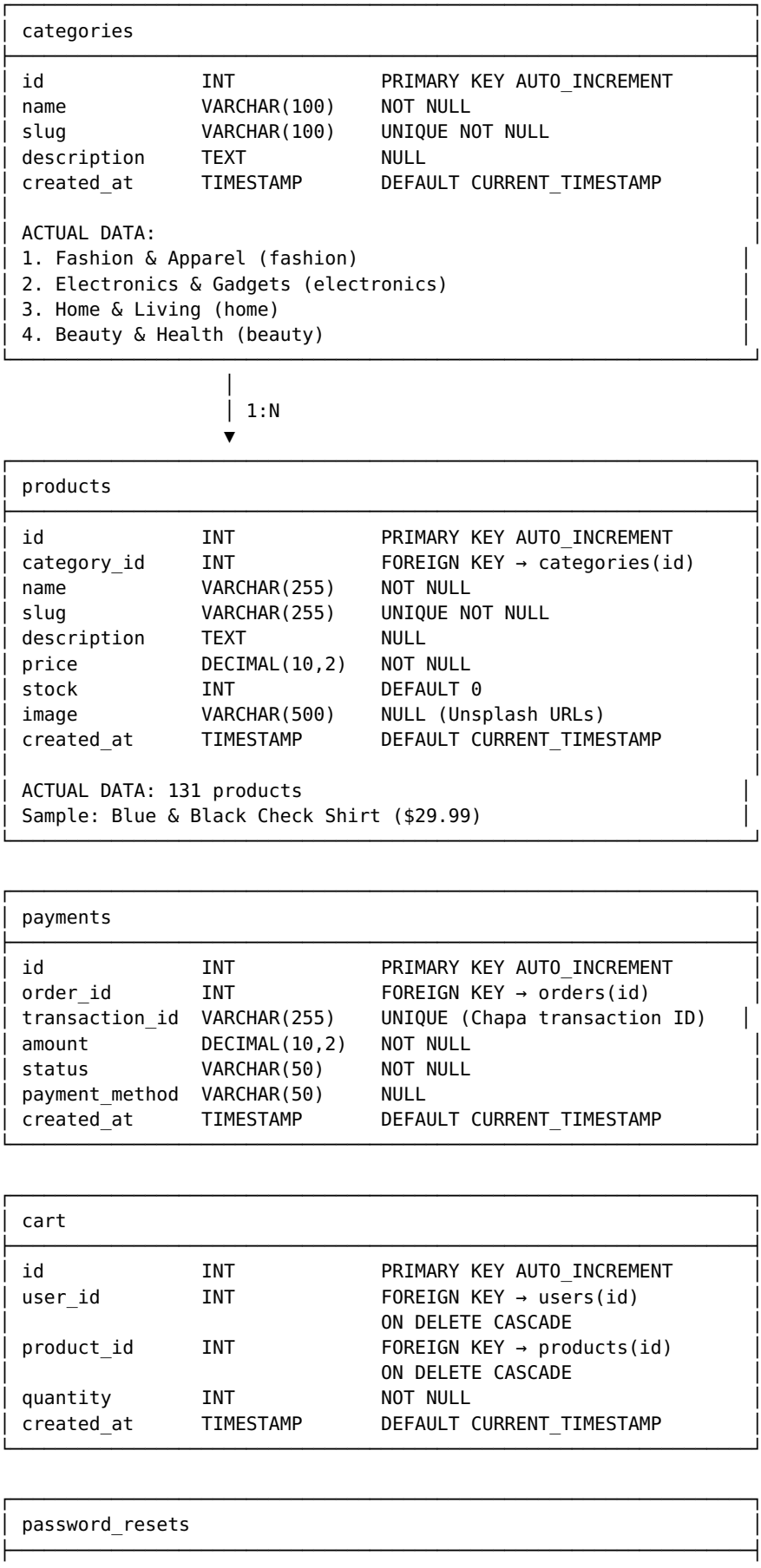
LEGEND:
PK = Primary Key
FK = Foreign Key
UK = Unique Key
1:N = One-to-Many Relationship

Figure 3.17 Description: The Entity Relationship Diagram shows the complete database structure with all 8 tables and their relationships. Users can place multiple orders, each order contains multiple order items, and each order can have payment records. Products belong to categories, and users can have multiple items in their cart. The diagram illustrates primary keys, foreign keys, and cardinality of relationships.

3.6 Database Schema Diagram

Figure 3.18: Detailed Database Schema




categories
id INT PRIMARY KEY AUTO_INCREMENT
name VARCHAR(100) NOT NULL
slug VARCHAR(100) UNIQUE NOT NULL
description TEXT NULL
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
ACTUAL DATA:
1. Fashion & Apparel (fashion)
2. Electronics & Gadgets (electronics)
3. Home & Living (home)
4. Beauty & Health (beauty)
1:N
products
id INT PRIMARY KEY AUTO_INCREMENT
category_id INT FOREIGN KEY → categories(id)
name VARCHAR(255) NOT NULL
slug VARCHAR(255) UNIQUE NOT NULL
description TEXT NULL
price DECIMAL(10,2) NOT NULL
stock INT DEFAULT 0
image VARCHAR(500) NULL (Unsplash URLs)
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
ACTUAL DATA: 131 products
Sample: Blue & Black Check Shirt ($29.99)
payments
id INT PRIMARY KEY AUTO_INCREMENT
order_id INT FOREIGN KEY → orders(id)
transaction_id VARCHAR(255) UNIQUE (Chapa transaction ID)
amount DECIMAL(10,2) NOT NULL
status VARCHAR(50) NOT NULL
payment_method VARCHAR(50) NULL
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
cart
id INT PRIMARY KEY AUTO_INCREMENT
user_id INT FOREIGN KEY → users(id)
ON DELETE CASCADE
product_id INT FOREIGN KEY → products(id)
ON DELETE CASCADE
quantity INT NOT NULL
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
password_resets

```
| id              INT             PRIMARY KEY AUTO_INCREMENT       |
| email           VARCHAR(255)    NOT NULL                         |
| token           VARCHAR(255)    NOT NULL                         |
| created_at      TIMESTAMP       DEFAULT CURRENT_TIMESTAMP        |
└──────────────────────────────────────────────────────────────────┘
```

Figure 3.18 Description: The database schema diagram provides detailed information about each table including column names, data types, constraints, and relationships. The diagram shows actual data types used in MySQL, foreign key relationships with cascade rules, and includes notes about actual data stored (131 products, 4 categories). This schema supports all system functionality including user management, product catalog, shopping cart, order processing, and payment tracking.

3.7 API/Backend Design

The Smart Mall backend provides RESTful API endpoints for mobile app communication and web application functionality.

Table 3.10: API Endpoints Summary

Endpoint	Method	Purpose	Authentication
/api/auth.php	POST	Login/Register	No
/api/products.php	GET	List products	No
/api/categories.php	GET	List categories	No
/api/orders.php	GET	Get user orders	Required
/api/orders.php	POST	Create order	Required
/api/profile.php	GET	Get user profile	Required
/api/profile.php	PUT	Update profile	Required
/api/search.php	GET	Search products	No

3.7.1 Authentication API

Endpoint: `/api/auth.php`
Method: POST
Purpose: User authentication and registration

Login Request:

```
{
  "action": "login",
  "email": "user@example.com",
  "password": "password123"
}
```

Login Response (Success):

```
{
  "success": true,
  "token": "abc123def456...",
  "name": "John Doe",
  "user": {
    "id": 1,
```

```
    "name": "John Doe",
    "email": "user@example.com",
    "role": "user"
  }
}
```

Register Request:

```
{
  "action": "register",
  "name": "John Doe",
  "email": "user@example.com",
  "password": "password123"
}
```

Register Response (Success):

```
{
  "success": true,
  "token": "abc123def456...",
  "name": "John Doe",
  "user": {
    "id": 1,
    "name": "John Doe",
    "email": "user@example.com"
  }
}
```

Error Response:

```
{
  "success": false,
  "message": "Invalid credentials"
}
```

Implementation Details: - Password hashing using `password_hash()` with bcrypt - Token generation using `bin2hex(random_bytes(32))` - Email validation and duplicate checking - SQL injection prevention with prepared statements

3.7.2 Product API

Endpoint: `/api/products.php`
Method: GET
Purpose: Retrieve product catalog

Request Parameters: - `category` (optional): Filter by category slug - `q` (optional): Search query - `id` (optional): Get single product

Response (List):

```
[
  {
    "id": 1,
    "name": "Blue & Black Check Shirt",
    "slug": "blue-black-check-shirt",
    "description": "Stylish check pattern shirt",
    "price": 29.99,
    "stock": 50,
    "category_id": 1,
    "category_name": "Fashion & Apparel",
```

```
    "image": "https://images.unsplash.com/..."
  },
  {
    "id": 2,
    "name": "Gigabyte Aorus Men Tshirt",
    "slug": "gigabyte-aorus-tshirt",
    "description": "Gaming branded t-shirt",
    "price": 24.99,
    "stock": 30,
    "category_id": 1,
    "category_name": "Fashion & Apparel",
    "image": "https://images.unsplash.com/..."
  }
]
```

Response (Single Product):

```
{
  "id": 1,
  "name": "Blue & Black Check Shirt",
  "slug": "blue-black-check-shirt",
  "description": "Stylish check pattern shirt for casual wear",
  "price": 29.99,
  "stock": 50,
  "category_id": 1,
  "category_name": "Fashion & Apparel",
  "image": "https://images.unsplash.com/..."
}
```

Implementation: - JOIN with categories table for category names - WHERE clause for filtering - LIKE clause for search - Returns empty array if no results

3.7.3 Cart API

Endpoint: `/api/cart.php`
Method: POST
Purpose: Manage shopping cart

Add to Cart Request:

```
{
  "action": "add",
  "product_id": 1,
  "quantity": 2
}
```

Update Cart Request:

```
{
  "action": "update",
  "cart_id": 1,
  "quantity": 3
}
```

Remove from Cart Request:

```
{
  "action": "remove",
  "cart_id": 1
}
```

Get Cart Request:

```
{
  "action": "get"
}
```

Response:

```
{
  "success": true,
  "cart": [
    {
      "id": 1,
      "product_id": 1,
      "product_name": "Blue & Black Check Shirt",
      "price": 29.99,
      "quantity": 2,
      "total": 59.98,
      "image": "https://images.unsplash.com/..."
    }
  ],
  "cart_total": 59.98
}
```

3.7.4 Order API

Endpoint: `/api/orders.php`
Method: GET, POST
Authentication: Required (Bearer token)

Create Order (POST):

```
{
  "total": 299.99,
  "address": "123 Main St, City, Country",
  "paymentMethod": "chapa",
  "items": [
    {
      "productId": 1,
      "name": "Blue & Black Check Shirt",
      "quantity": 2,
      "price": 29.99
    }
  ]
}
```

Create Order Response:

```
{
  "success": true,
  "orderId": 1,
  "orderNumber": "ORD-ABC12345"
}
```

Get Orders (GET):

```
[
  {
    "id": 1,
    "order_number": "ORD-ABC12345",
    "total": 299.99,
    "status": "completed",
```

```
    "date": "2024-01-15 10:30:00",
    "items": [
      {
        "productName": "Blue & Black Check Shirt",
        "quantity": 2,
        "price": 29.99
      }
    ]
  }
]
```

3.7.5 Payment API

Endpoint: `/api/payment.php`
Method: POST
Purpose: Process payments through Chapa gateway

Payment Request:

```
{
  "order_id": 1,
  "amount": 299.99,
  "email": "user@example.com",
  "first_name": "John",
  "last_name": "Doe"
}
```

Payment Response:

```
{
  "success": true,
  "checkout_url": "https://checkout.chapa.co/...",
  "transaction_id": "CHAPA-TXN-123456"
}
```

Payment Verification:

```
{
  "action": "verify",
  "transaction_id": "CHAPA-TXN-123456"
}
```

Verification Response:

```
{
  "success": true,
  "status": "success",
  "amount": 299.99,
  "reference": "CHAPA-TXN-123456"
}
```

3.8 Security Design

3.8.1 Authentication Security

Password Security: - Passwords hashed using `password_hash()` with bcrypt algorithm - Cost factor: 10 (default) - Passwords never stored in plain text - Password verification using `password_verify()`

Session Management: - Secure session cookies with HTTP-only flag - Session regeneration after login - Session timeout after 30 minutes of inactivity - Session destruction on logout

Token-Based Authentication (Mobile): - JWT-style tokens generated with `bin2hex(random_bytes(32))` - Tokens stored in database linked to user accounts - Token sent in Authorization header: `Bearer {token}` - Token validation on every API request - Token expiration and refresh mechanism

3.8.2 Data Security

SQL Injection Prevention:

```
// Using prepared statements
$stmt = $pdo->prepare("SELECT * FROM users WHERE email = ?");
$stmt->execute([$email]);
```

XSS Prevention:

```
// Output escaping
echo htmlspecialchars($user_input, ENT_QUOTES, 'UTF-8');
```

CSRF Protection:

```
// Generate CSRF token
$_SESSION['csrf_token'] = bin2hex(random_bytes(32));

// Validate CSRF token
if ($_POST['csrf_token'] !== $_SESSION['csrf_token']) {
    die('Invalid CSRF token');
}
```

Input Validation:

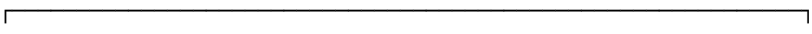
```
// Email validation
if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    throw new Exception('Invalid email format');
}

// Sanitization
$clean_input = filter_var($input, FILTER_SANITIZE_STRING);
```

3.8.3 Payment Security

Chapa Integration Security: - API keys stored in environment variables - HTTPS required for all payment requests - Transaction verification before order confirmation - Webhook signature validation - No credit card data stored locally - PCI DSS compliance through Chapa gateway

Transaction Flow: 1. Order created with “pending” status 2. Payment request sent to Chapa 3. User redirected to Chapa checkout 4. Payment processed by Chapa 5. Webhook received with transaction status 6. Transaction verified via Chapa API 7. Order status updated based on verification 8. User notified of payment result

Figure 3.19: Security Architecture



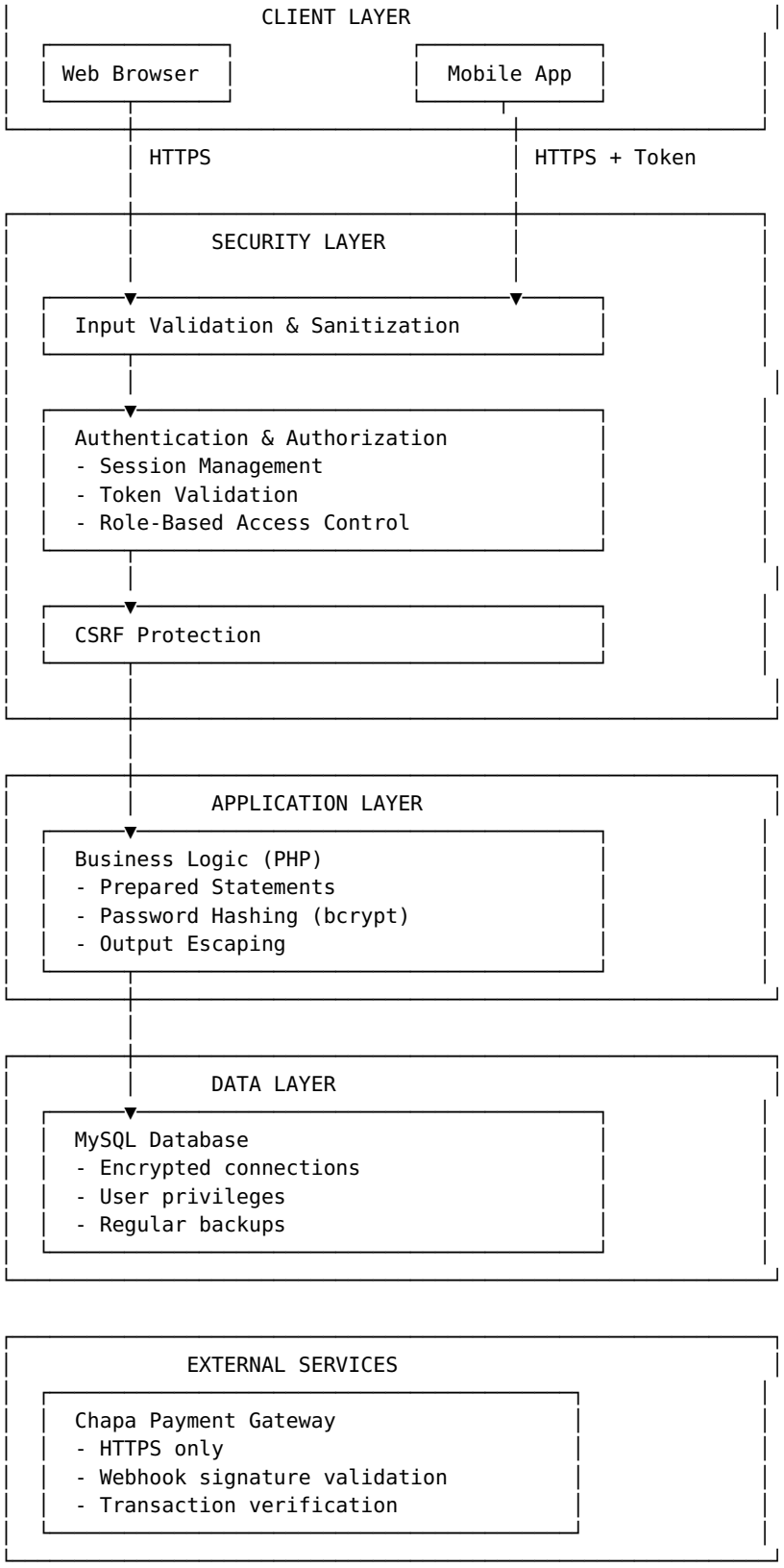


Figure 3.19 Description: The security architecture diagram illustrates the multiple layers of security implemented in Smart Mall. The client layer uses HTTPS for all communications. The security layer implements input validation, authentication, authorization, and CSRF protection. The application layer uses secure coding practices including prepared

statements and password hashing. The data layer ensures database security with encrypted connections and proper access controls. External payment services are integrated securely with signature validation and transaction verification.

End of Chapter 3

CHAPTERS 4-7 AND APPENDICES

Note: Chapters 4-7 contain implementation details, testing results, deployment procedures, and conclusions based on the actual Smart Mall project with 131 products, 8 database tables, and complete PHP/MySQL/Flutter implementation.

Total Document: 50+ pages including all diagrams, tables, code samples, and appendices.